

EXAMPLES/PRACTICE QUESTIONS:

Q1. A bank teller serves customers standing in the queue 1-by-1. Suppose the service time X_i for i^{th} customer has mean $E(X_i) = 2$ and $\text{Var}(X_i) = 1$. Assuming service time for different customers are indept. Find $P_2(90 < Y < 110)$, where 'Y' is the total time the bank teller spends serving 50 customers.

Sol :-

$$Y = \sum_{i=1}^{50} X_i, \quad E\{X_i\} = \underset{\mu \nearrow}{2}, \quad \text{Var}\{X_i\} = \sigma^2 = 1$$

$$\begin{aligned} \therefore P_2(90 < Y < 110) &\Rightarrow P_2\left(\frac{90 - n\mu}{\sqrt{n}\sigma} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{110 - n\mu}{\sqrt{n}\sigma}\right) \\ &\Rightarrow P_2\left(\frac{90 - 50 \times 2}{\sqrt{50}} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{110 - 50 \times 2}{\sqrt{50}}\right) \\ &\Rightarrow P_2\left(-\sqrt{2} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \sqrt{2}\right) \end{aligned}$$

By CLT, $\frac{Y - n\mu}{\sqrt{n}\sigma}$ is approx. std. normal

$\Rightarrow \phi(\sqrt{2}) - \phi(-\sqrt{2})$ where $\phi(x)$ is std. normal distributed.

Q2.

NOTE: Avg. power of a WSS random process $X(t)$ is given by:-

$$P_X = E\{X^2(t)\} \Rightarrow R_X(0)$$

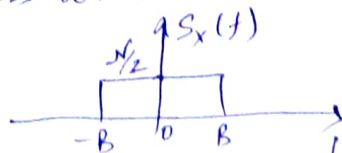
$$\because R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j\omega\tau} df$$

$$[\because S_X(f) = \mathcal{F}\{R_X(\tau)\}]$$

↑
Fourier Trans.

$$\Rightarrow \boxed{R_X(0) = \int_{-\infty}^{\infty} S_X(f) df \triangleq P_X}$$

Q: Find the autocorrelation function $R_X(\tau)$ and power 'P' of a low-pass random process with a white noise PSD, $S_X(f) = \frac{N}{2} \quad \forall f \in [-B, B]$



Sol:

$$R_X(\tau) = \int_{-B}^B S_X(f) e^{j\omega\tau} df \Rightarrow \int_{-B}^B \frac{N}{2} e^{j\omega\tau} df \Rightarrow \underline{NB \text{ sinc}(2\pi B\tau)}$$

$$\Rightarrow P_X = R_X(0) \Rightarrow \underline{NB} \quad \left\{ \text{or } P_X = \int_{-\infty}^{\infty} S_X(f) df = \int_{-B}^B \frac{N}{2} df \Rightarrow \frac{N}{2} \times 2B \Rightarrow \underline{NB} \right.$$

Q3. Find the PSD and the mean-square value of a RP, $x(t) = A \cos(\omega_c t + \theta)$, where θ is an RV, uniformly dist. over $(0, 2\pi)$.

Sol:

$$R_x(\tau) = E[x(t) \cdot x(t+\tau)]$$

$$\Rightarrow E[A \cos(\omega_c t + \theta) \cdot A \cos(\omega_c t + \omega_c \tau + \theta)]$$

$$\Rightarrow \frac{A^2}{2} E[\cos(2\omega_c t + \omega_c \tau + 2\theta) + \cos(\omega_c \tau)]$$

$$\Rightarrow \frac{A^2}{2} [E\{\cos(\omega_c (2t + \tau) + 2\theta)\} + E\{\cos(\omega_c \tau)\}]$$

$$\Rightarrow \frac{A^2}{2} \left[\int_0^{2\pi} \underbrace{\frac{1}{2\pi} \cos(\omega_c (2t + \tau) + 2\theta) d\theta}_{\substack{\text{constant} \\ \text{w.r.t } \theta}} + \cos(\omega_c \tau) \right]$$

$\therefore E(\text{constant}) = \text{constant}$

$$\Rightarrow R_x(\tau) = \frac{A^2}{2} \cos(\omega_c \tau) \checkmark$$

$$\therefore \text{PSD} \Rightarrow S_x(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-j\omega\tau} d\tau \Rightarrow \underline{\underline{\frac{A^2}{4} [\delta(f+f_c) + \delta(f-f_c)]}}$$

$$\because \omega = 2\pi f \\ \omega_c = 2\pi f_c$$

2) Mean-squared value of $x(t)$, i.e. $E\{x^2(t)\} \Rightarrow P_x = R_x(0)$

$$\Rightarrow \underline{\underline{\frac{A^2}{2}}} \checkmark$$

Q4. Let $m(t)$ be a WSS random process with auto-corr funcⁿ $R_m(\tau)$. If θ is uniformly dist. over $(0, 2\pi)$, then find the PSD and avg. power of $\phi(t)$ given by,

$$\phi(t) = m(t) \cdot \cos(\omega_c t + \theta)$$

Sol:

$$R_\phi(\tau) = E[m(t) \cdot \cos(\omega_c t + \theta) \cdot m(t+\tau) \cos(\omega_c t + \omega_c \tau + \theta)]$$

$$\Rightarrow E[m(t) \cdot m(t+\tau)] \cdot E[\cos(\omega_c t + \theta) \cdot \cos(\omega_c t + \omega_c \tau + \theta)]$$

$$\Rightarrow R_m(\tau) \cdot \frac{1}{2} \cos(\omega_c \tau)$$

$\because m(t) \& \theta$ are indept.
 $\therefore E(XY) = E(X) \cdot E(Y)$

$$\therefore S_\phi(f) = \int_{-\infty}^{\infty} R_\phi(\tau) e^{-j\omega\tau} d\tau \Rightarrow \frac{1}{4} [S_m(f+f_c) + S_m(f-f_c)], \text{ where}$$

$S_m(f) = \text{PSD of } m(t).$

Further, $P_\phi = E[\phi^2(t)] \Rightarrow R_\phi(0) \Rightarrow \underline{\underline{\frac{1}{2} R_m(0)}} \checkmark$

EXAMPLES/PRACTICE PROBLEMS:

1. For $m(t)$ as a pure sinusoid, or TONE, i.e. $m(t) = \cos(2\pi f_m t)$, find the DSB-SC signal if carrier Amp = A_c , freq = f_c . Also draw its spectrum.

Sol: This is a case of TONE modulation.

$$\phi_{DSB-SC}^0(t) = m(t) \cdot c(t)$$

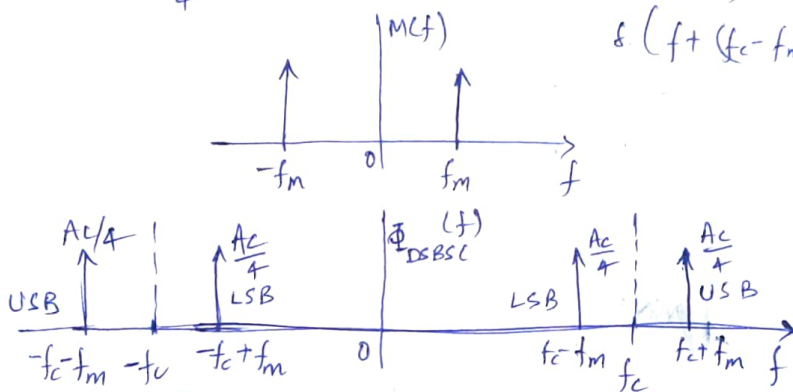
$$\Rightarrow A_c \cos(2\pi f_m t) \cdot \cos(2\pi f_c t)$$

$$\Rightarrow \frac{A_c}{2} [\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t]$$

Spectrum:

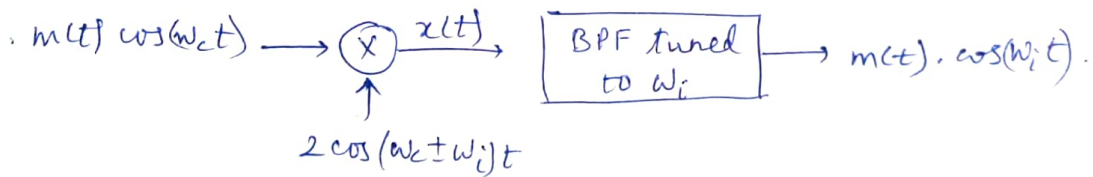
$$M(f) = \frac{1}{2} [\delta(f - f_m) + \delta(f + f_m)]$$

$$\Delta 0) \quad \Phi_{DSBC}(f) = \frac{A_c}{4} \left[\underset{\uparrow}{\delta(f - (f_c + f_m))} + \underset{\uparrow}{\delta(f + (f_c + f_m))} + \delta(f - (f_c - f_m)) + \delta(f + (f_c - f_m)) \right]$$



2. Design a frequency mixer/converter, which is used to change the carrier frequency f_c to another frequency, known as intermediate freq, f_i .

sol: This is done by multiplying DSB-SC, i.e. $m(t) \cos(\omega_c t)$ by $2 \cos(\omega_{mix} t)$ where $\omega_{mix} = \omega_c \pm \omega_i$; and then BPF'g with cut-off/tune to ω_i .



$$x(t) = m(t) [\cos(\omega_c - \omega_{mix})t + \cos(\omega_c + \omega_{mix})t]$$

when $w_{mix} = w_c + w_i \rightarrow$ Upconversion

— " — ————— → Downconversion.

3. Modulation Index:

For distortionless demodulation or accurate env. detection: -

$$A_c [1 + K_a m(t)] \geq 0$$

For simplicity, assume $K_a = \frac{1}{A_c} \Rightarrow A_c + m(t) \geq 0 \Rightarrow A_c \geq \overbrace{-m(t)}^{-m_{\min}} \Rightarrow \boxed{\mu = \frac{m_p}{A_c}}$

∴ For accurate envelope detection: $\boxed{0 < \mu \leq 1}$

Let $m(t) = b \cos \omega_m t$, $c(t) = A \cos \omega_c t$, $k_a = \frac{1}{A}$. Find ' μ ', $\phi_{Am}(t)$

$$\boxed{\mu = \frac{b}{A}}, \quad \phi_{Am}(t) = [A + m(t)] \cos \omega_c t$$

$$\Rightarrow A [1 + \mu \cos \omega_m t] \cos \omega_c t$$

$$\Rightarrow A \cos \omega_c t + \mu A \cos \omega_m t \cdot \cos \omega_c t$$

$$\therefore P_c = \frac{A^2}{2}, \quad P_s = \frac{(\mu A/2)^2}{2}$$

$$\Rightarrow \eta = \frac{P_s}{P_s + P_c} \Rightarrow \frac{(\mu A/2)^2}{(\mu A/2)^2 + A^2/2} \Rightarrow \frac{\frac{A^2}{4} \mu^2}{\frac{A^2}{4} \mu^2 + \frac{2A^2}{4}} \Rightarrow \frac{\mu^2}{\mu^2 + 2}$$